Table 1: One-pass residual readout vs. fairly-conditioned autoregressive loop (accuracy). Readout = small MLP on the frozen final residual (4-seed vote); loop = next-token log-prob over the closed answer set, full context, balanced few-shot. Δ = readout $-$ loop.k, paired McNemar on identical rows ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). All runs batch 8; Qwen3 0.6B/4B on H200, Qwen3-8B/Mistral-7B/Granite-3.1-8B on B200, DeepSeek-V4-Flash on B300. [a] BoolQ loops: Qwen at pad 8192 with best $k \in \{0, \ldots, 64\}$, Mistral/Granite/DSV4 at pad 2048, k=8. [b] RuleTaker n2k: readout scored on the loop's exact 400 rows. [c] ARC-Challenge: full test (1165 rows), k=8, pad 2048.

Task	Model	Readout	loop.0	loop.k	Δ	p (McNemar)
BoolQ[a]	Qwen3-0.6B	**0.753**	0.631	0.715	**+0.038*****	2.4e$-$05
	Qwen3-4B	0.862	0.854	**0.869**	-0.007	0.22
	Qwen3-8B	0.879	0.862	**0.886**	-0.007	0.24
	Mistral-7B	0.841	0.798	**0.852**	-0.011	0.21
	Granite-3.1-8B	0.854	0.815	**0.864**	-0.010	0.64
	DeepSeek-V4-Flash	0.896	0.888	**0.906**	-0.009	0.39
RuleTaker n2k[b]	Qwen3-0.6B	**0.638**	0.600	0.545	**+0.093****	3.7e$-$03
	Qwen3-4B	**0.738**	0.653	0.698	$+0.040$	0.20
	Qwen3-8B	**0.753**	0.675	0.730	$+0.023$	0.45
	Mistral-7B	**0.660**	0.515	0.575	**+0.085***	0.017
	Granite-3.1-8B	**0.828**	0.558	0.695	**+0.133*****	1.3e$-$06
	DeepSeek-V4-Flash	0.763	0.605	**0.838**	-0.074^{**}	1e$-$03
ARC-Challenge[c]	Qwen3-0.6B	0.492	0.502	**0.597**	-0.106^{***}	3.2e$-$11
	Qwen3-4B	0.837	0.826	**0.882**	-0.046^{***}	2.9e$-$06
	Qwen3-8B	0.908	0.902	**0.913**	-0.005	0.57
	Mistral-7B	0.728	0.750	**0.779**	-0.051^{***}	9.3e$-$05
	Granite-3.1-8B	0.708	0.732	**0.790**	-0.082^{***}	5.8e$-$10
	DeepSeek-V4-Flash	0.951	0.961	**0.959**	-0.007	0.23

Table 2: Readout probe-placement sweeps (18 cells, all batch 8): best mid-depth tap vs the final-layer readout, paired McNemar on identical rows. *Layers* = stack depth (28/36/36/32/40/43); taps are residual-layer indices. *Plateau onset* = first swept layer within 1 pt of the final-layer readout — the verdict representation is fully assembled from there on; at DeepSeek-V4 this happens by L22/42 (52% depth) on *all three tasks*, with ARC showing the sharpest snap-in (near-chance at L15 → 0.944 at L22). Mid-depth taps win significantly at 9/18 cells and never lose significantly.

Task	Model	Layers	Best tap	Depth	Plateau onset	mlp (best)	mlp (final)	Δ	p
BoolQ	Qwen3-0.6B	28	L18	64%	L18 (67%)	0.761	0.746	+0.010	0.18
	Qwen3-4B	36	L24	67%	L24 (69%)	0.868	0.856	**+0.011***	0.038
	Qwen3-8B	36	L30	83%	L24 (69%)	0.889	0.878	**+0.011***	0.016
	Mistral-7B	32	L16	50%	**L16 (52%)**	0.863	0.841	**+0.022*****	1.1e−04
	Granite-3.1-8B	40	L20	51%	**L17 (44%)**	0.868	0.854	**+0.018****	1.0e−03
	DeepSeek-V4-Flash	43	L36	86%	**L22 (52%)**	0.900	0.896	+0.005	0.20
RuleTaker n2k	Qwen3-0.6B	28	L18	64%	**L13 (48%)**	0.688	0.650	**+0.036***	0.020
	Qwen3-4B	36	L24	67%	L24 (69%)	0.781	0.744	**+0.038****	2.8e−03
	Qwen3-8B	36	L24	67%	L24 (69%)	0.778	0.758	**+0.025***	0.040
	Mistral-7B	32	L18	56%	**L11 (35%)**	0.702	0.643	**+0.059*****	5.2e−05
	Granite-3.1-8B	40	L20	51%	L20 (51%)	0.816	0.805	+0.011	0.37
	DeepSeek-V4-Flash	43	L29	67%	**L22 (52%)**	0.776	0.765	+0.021	0.057
ARC-Challenge	Qwen3-0.6B	28	L26	93%	L26 (96%)	0.500	0.492	+0.013	0.082
	Qwen3-4B	36	L29	81%	L24 (69%)	0.856	0.841	**+0.022*****	2.2e−04
	Qwen3-8B	36	L35	97%	L24 (69%)	0.909	0.909	+0.001	1.00
	Mistral-7B	32	L20	63%	L18 (58%)	0.741	0.728	+0.013	0.50
	Granite-3.1-8B	40	L39	100%	L39 (100%)	0.708	0.708	0.000	—
	DeepSeek-V4-Flash	43	L42	98%	**L22 (52%)**	0.951	0.951	−0.005	0.29

Table 3: What each access to the frozen residual costs (economics orientation, not a benchmark). The point is the *type* of cost, and that the readout is the only access that both fits one forward pass *and* never generates. Status stamps every row: "measured" = our run protocol; "projected" = forward arithmetic we did not run; "off-axis" = a real competing claim on a different cost axis (memory bandwidth), not operator FLOPs. Readout & scoring-loop are the same single pass at equal context — the readout's structural gap is **no generation**, not layer-skipping. Early-exit serving (EE-LLM, CALM, LayerSkip, layer-removal) shares our placement premise and is prior art for any "skip the tail" claim; it is not claimed here. Operator-FLOP figures would need true (active) parameter counts per model and a per-cell context length — footnoted, not fabricated.

Access	Cost shape	Generates?	Status
readout (final tap)	1 fwd , short ctx	no	measured
scoring loop ".8"	1 fwd, long ctx	no	measured
early-exit decode	1 fwd, tail truncated	yes	projected / prior art
decode / output	K fwds (per token)	yes	projected

Footnote. "Operator-FLOP" = total multiply-add throughput, not wall-clock; memory-bandwidth engineering is a separate axis. Any absolute FLOPs number would require each model's *active* parameter count (routed, for MoE) and its per-cell context length (pad_max); both are open (DSV4 active-params, and no self-terminating decode was run).